

QO-100 quick-start (sdroxide + ADALM-Pluto)

A task-oriented walkthrough for getting an ADALM-Pluto + Ku-band LNB station onto the QO-100 (Es'hail-2) geostationary transponder with sdroxide. It covers the one-time station setup, locking onto the bird, and the QO-100 beacon-sync tab. For the reference detail behind each control, see [USER MANUAL.md](#) §2.21 (QO-100 beacon plugin) and §6.2.7 (PlutoSDR).

The 10 GHz downlink comes down through the LNB; the 2.4 GHz uplink leaves the Pluto directly.

What you need

- ADALM-Pluto (or a LibreSDR / ANTSDR-class AD936x board).
- A universal Ku-band LNB with a 9750 MHz local oscillator — the usual QO-100 choice.
- A dish, the coax from the LNB, and a 2.4 GHz uplink antenna.
- sdroxide installed. The Pluto driver is built in; there is nothing else to install.

The frequency chain

```
downlink    10489.750 MHz --> LNB (LO 9750 MHz) --> 739.750 MHz --> Pluto
uplink      2400.xxx MHz <-- Pluto (direct, not converted)
```

Bold names such as **Converter** are buttons and fields exactly as they appear on screen. `Settings > Radio` is a menu path; `192.168.2.1` is a value you type.

Part A — Settings

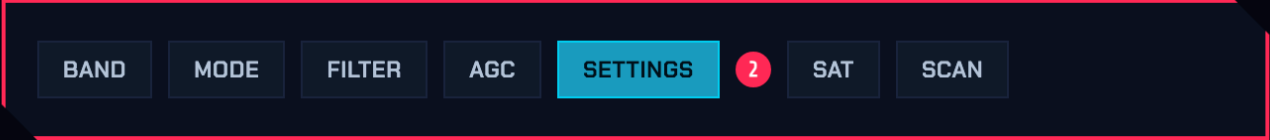
Done once and saved. Everything is in the **Settings** window, on the **General** and **Radio** tabs.

1. Start sdroxide

Open the program. The main window has the control bar across the top and the panadapter and waterfall below it.

2. Open the settings

Press **SETTINGS** on the top control bar.



A horizontal control bar with seven buttons: BAND, MODE, FILTER, AGC, SETTINGS, SAT, and SCAN. The SETTINGS button is highlighted with a red circle containing the number 2. The entire bar is enclosed in a red rectangular frame.

3. Callsign and locator

On the **General** tab, enter your callsign in **Callsign** and your Maidenhead grid (e.g. `KN41GG`) in **Locator**. The satellite lock and the world map use this locator — **LOCK ON** will not work without it.

Settings

General

Radio

Servers

TLE

Callsign

TA1XYZ

3

Locator

KN41GG

3

4. Radio tab – device: PlutoSDR

Switch to the **Radio** tab and pick **PlutoSDR** in the interface selector. A Pluto is a network device even on a USB cable: plugging it in creates a network adapter, not a serial port. The next five fields are on this same tab:

Settings

General

Radio

Servers

Interface

PlutoSDR

4

Converter

LNB, Ku low (-9750 MHz)

5

Offset

-9 750 000 000 Hz

[from preset, auto]

Transmit

Its own offset

0

Hz

6

Address

192.168.2.1

Discover

7

Sample rate

1 Msps



Full duplex

8

Apply

9

5. Converter: LNB, Ku low (-9750 MHz)

In the **Converter** dropdown, pick the **LNB, Ku low [-9750 MHz]** preset — because our LNBs have a 9750 MHz local oscillator. This fills in the `-9 750 000 000` Hz offset below automatically, so the 10489.750 MHz beacon lands near the right place on the dial.

6. Transmit row: Its own offset = 0

In the **Transmit** row next to the converter offset, choose **Its own offset** and type `0` into the Hz box beside it.

There is no need to enter a TX figure such as 8085 MHz the way we used to. This `0` means "the downlink comes through the LNB, but the 2.4 GHz uplink leaves the radio directly — the transmit side is not converted". If you later find your TX frequency is off, correct it from this box (in Hz, same sign rule as the receive offset).

7. Pluto address

In the connection section below, type the Pluto's IP into **Address**, or let **Discover** find it. Over USB, `192.168.2.1` is enough. On a LAN, use the Pluto's address on that network. `ip:192.168.2.1` is also accepted; an address starting with `usb:` is refused, because this backend reaches the radio over the network the USB cable already provides.

8. Sample rate and Full duplex

Set **Sample rate** to `1 Msps` and tick **Full duplex** below it, so you keep hearing your own downlink while transmitting.

Note. **Full duplex** only works over real Ethernet — a Pluto or LibreSDR behind a gigabit adapter. Over USB the station is half-duplex and you will not hear your own downlink during an over. Also, a stock Pluto cannot go below about 2.084 Msps; a lower value is rounded up and the connection message says so.

9. Apply and close

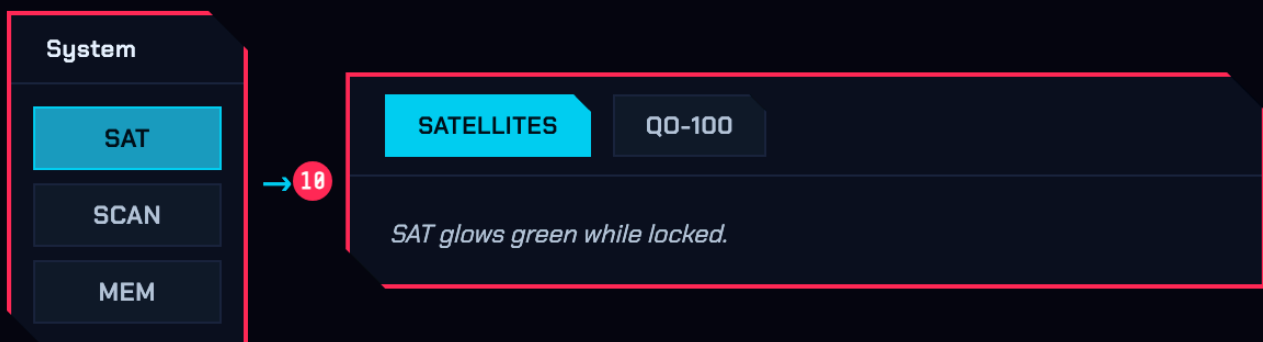
Press **Apply** at the bottom and close the **Settings** window. Settings take effect on Apply / reconnect.

Part B — Locking onto the satellite

Done each time you work the bird.

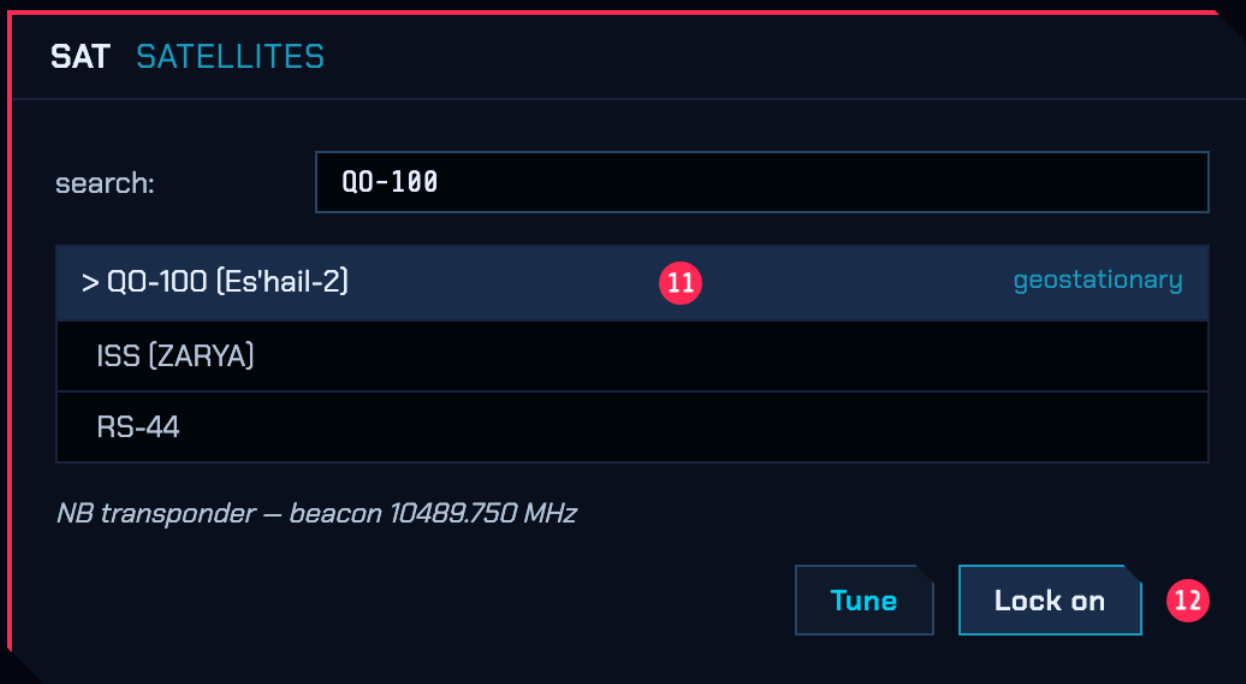
10. Open the SAT window

Press **SAT** in the System box. The button glows green while a lock is running; the correction keeps being applied even with the window closed.



11. Pick QO-100

On the **SATELLITES** tab, type **QO-100** into the search box and pick it from the list. Its published links (narrowband transponder, beacon) appear below.



12. LOCK ON

Press **LOCK ON** at the bottom. QO-100 is geostationary, so it needs no Doppler correction; the transponder mapping derives the 2.4 GHz uplink from wherever you set the downlink dial.

Part C — Beacon sync

Optional but recommended. If your LNB is not very stable, the QO-100 beacon-sync plugin (contributed by the ATÖLYE group) measures where the 10489.750 MHz narrowband beacon really is, corrects the converter/LNB offset so the dial and the signal agree, and keeps correcting it as the LNB warms up and drifts.

13. Switch to the QO-100 tab

It is the second tab of the SAT window.

Prerequisite. The plugin corrects a *residual* error of a few kHz to tens of kHz — it does not guess from scratch. The **Settings > Radio** converter offset has to be roughly right first (step 5), so the beacon lands somewhere near 10489.750 MHz.

SATELLITES

QO-100

ON

TELEMETRY

AUTO

width - ±25 kHz +



mini waterfall — hatched = park lane

target = 10489.750 MHz · beacon, two lobes

double-click the beacon's centre to mark it

TRACKER

CONVERTER OFFSET

MEASURED

+1.2 kHz (null 12 dB · snr 15 dB)

-9 749 920 000 Hz

10489.751200 MHz DRIFT +1.2 kHz

Apply correction

14. See the beacon in the strip

If the beacon is not visible as a band / two symmetric lobes, step **width** up with + (±5 kHz steps, up to ±50 kHz). Widen only until the two lobes and the null between them are clear, then bring it back toward ±5 kHz.

15. Still nothing: nudge 9750 by hand

Nudge the 9750 MHz converter offset in **Settings > Radio** to an approximate value in the right direction and press **Apply**; repeat until the beacon is inside the strip. The plugin takes it from there.

16. Simple (one-shot) correction

Double-click the middle of the beacon in the strip. That plants a "the beacon is here" mark (the two lobes, centred on the null). Then press **APPLY CORRECTION** at the bottom. The receiver reopens on the corrected offset and the beacon jumps toward the centre. Two passes — one wide (coarse), one narrow (fine) — get it within a few hundred Hz.

17. Automatic, continuous correction

Press **ON** (the spectral tracker starts — it only *measures*, it changes nothing), then **AUTO** (the loop closes: every clean, steady measurement is applied as a slow, rate-limited nudge to the offset).

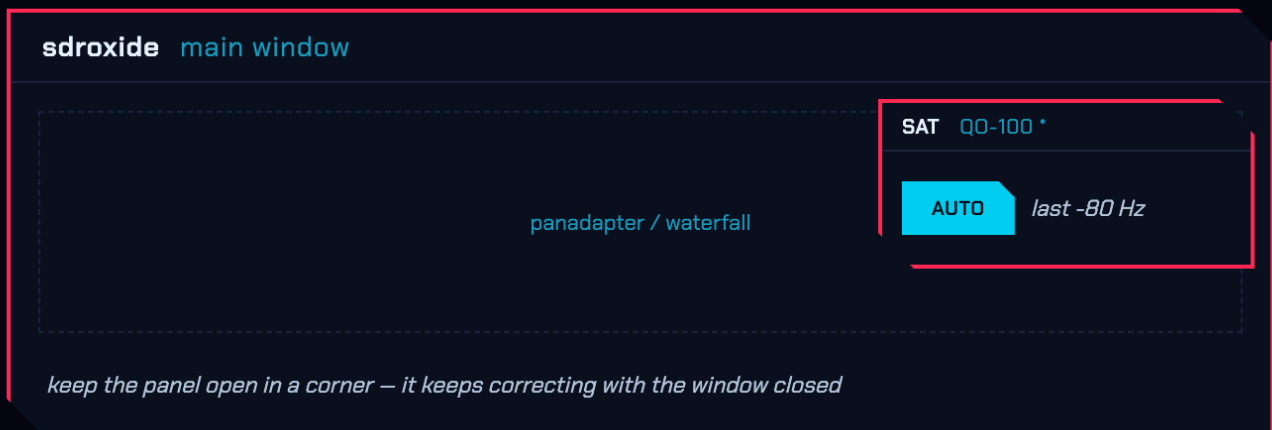
AUTO then corrects LNB drift in the background. A single noisy reading never yanks the receiver; it takes a run of agreeing measurements to move the offset. No correction is made *while you are transmitting* — each one reopens the receiver, which would cut off your over — so a held-back correction goes out as soon as the over ends. AUTO also watches its own work and switches itself off if the offset it writes does not move the beacon, or if it has gone further than any LNB could plausibly be out by, with a notice saying which.

18. Telemetry (optional)

Press **TELEMETRY** to run the AO-40 frame decoders; the beacon's own status text appears when it locks. Not needed for calibration — it is an independent check.

19. Leave it running in the background

When you are done, minimise this window and keep it in a corner of the screen (closing it is fine too). The LNB correction keeps going with the window closed; the **SAT** chip stays lit and the QO-100 tab keeps its dot.



Good QSOs, and 73.